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GOSIM Shenzhen 2026

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Beyond Voice

Building Voice-Native Agents

From Enterprise Lessons to the Next Generation of Voice AI Applications

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01

Voice as an interface to AI



AI agent adoption is accelerating

Voice Agent are at the forefront

2.45B
Active AI agents

by 2030

~80× growth from 2025

IDC

$43.8B
Global Voice AI Agents market

by 2036

24% CAGR from $5.1B in 2026

Meticulous RESEARCH
WHERE VALUE EXCEEDS PRICE

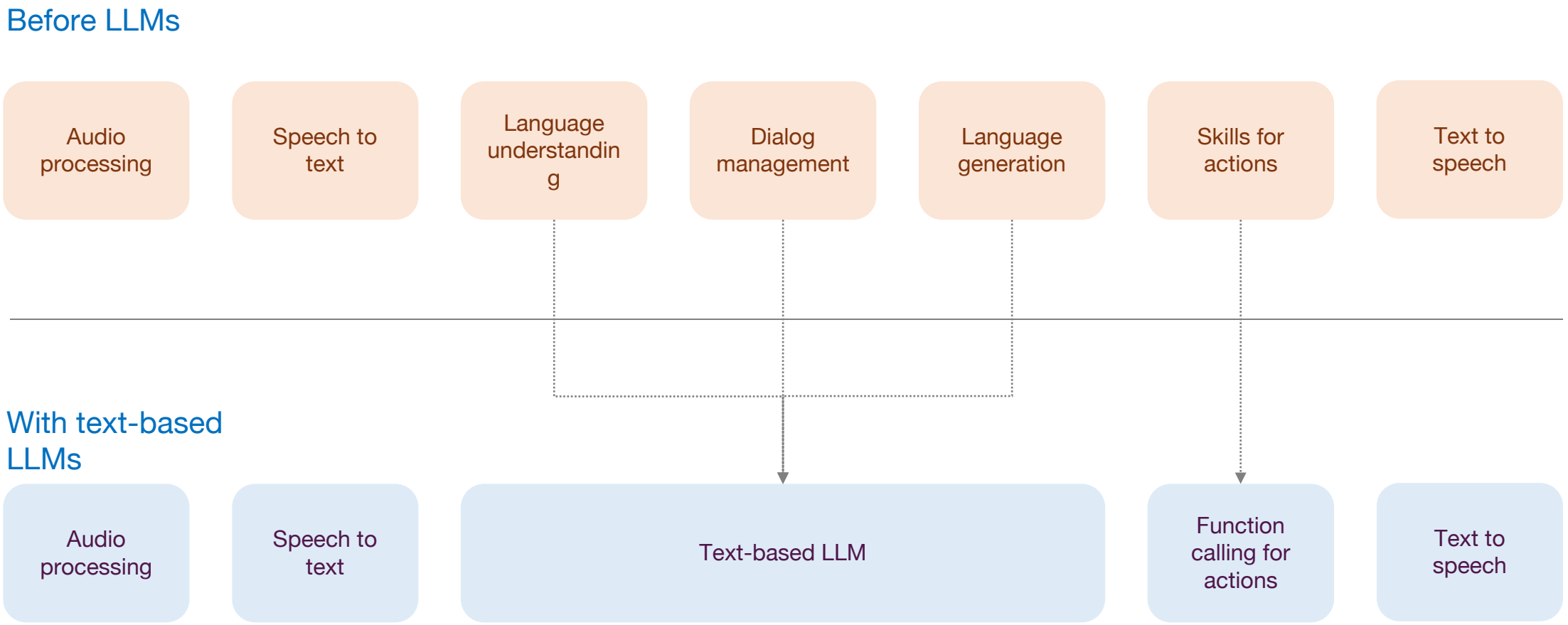
82%
of organizations intend to integrate them within 1–3 years

Capgemini

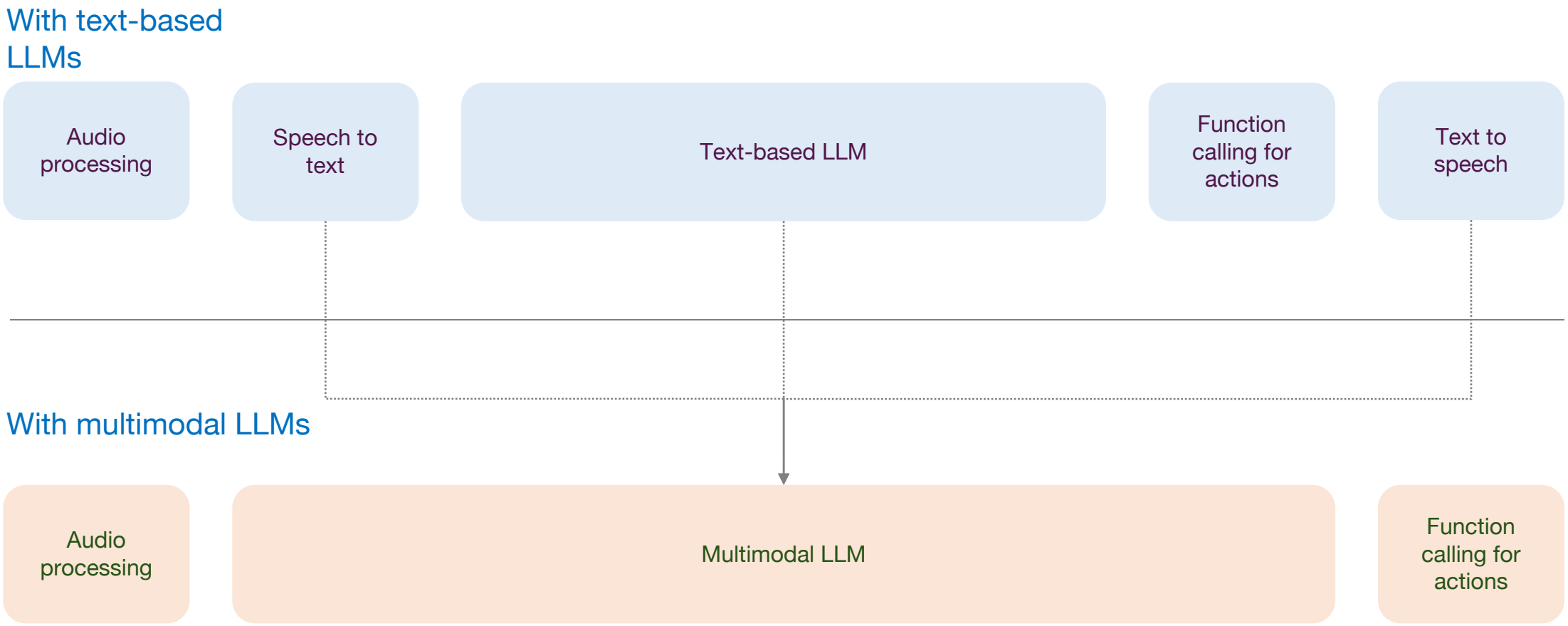
50%
of enterprises run voice AI agents in production in at least one business area in 2026.

IDC

Building voice agents is becoming easier



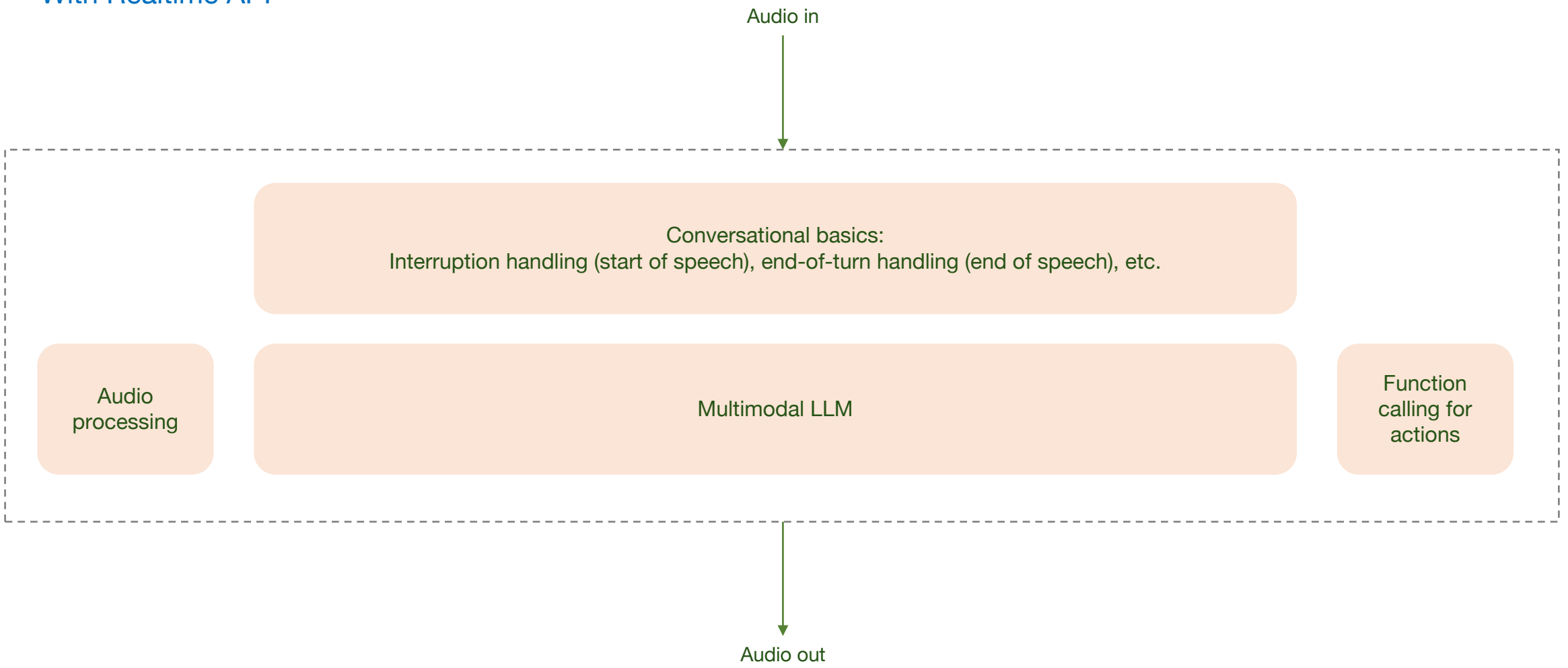
Building voice agents is becoming easier



Building voice agents is becoming easier

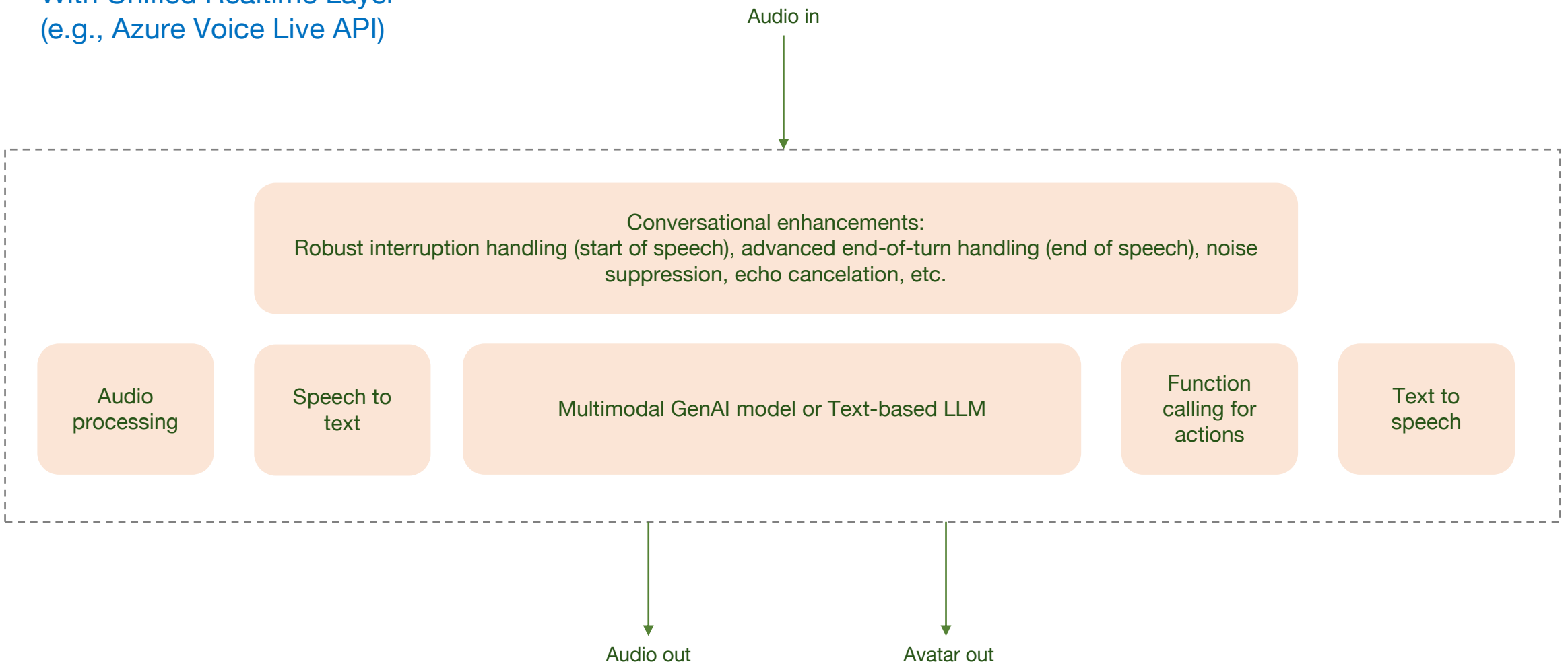
Previously,
multi-requests
turn-by-turn

With Realtime API



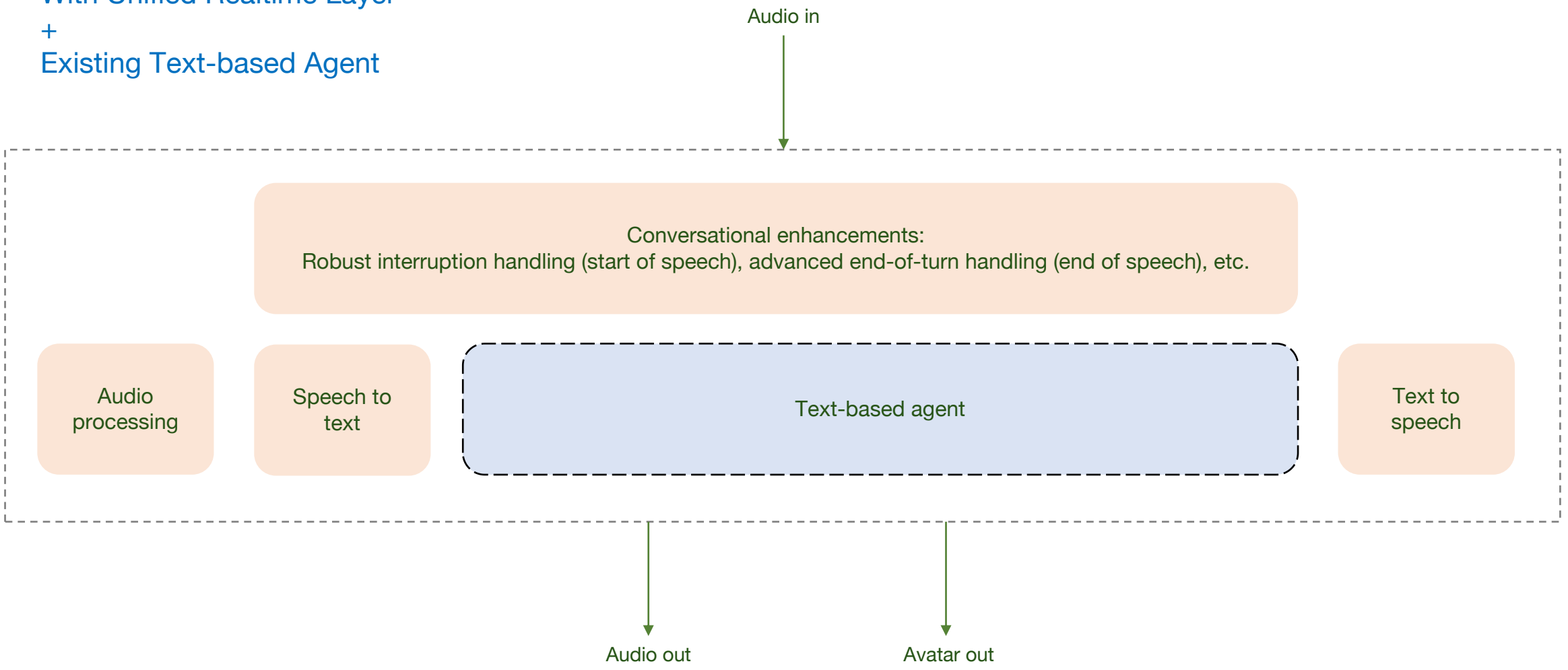
Building voice agents is becoming easier

With Unified Realtime Layer
(e.g., Azure Voice Live API)



Building voice agents is becoming easier

With Unified Realtime Layer
+
Existing Text-based Agent



But

It’s harder to build voice experience well

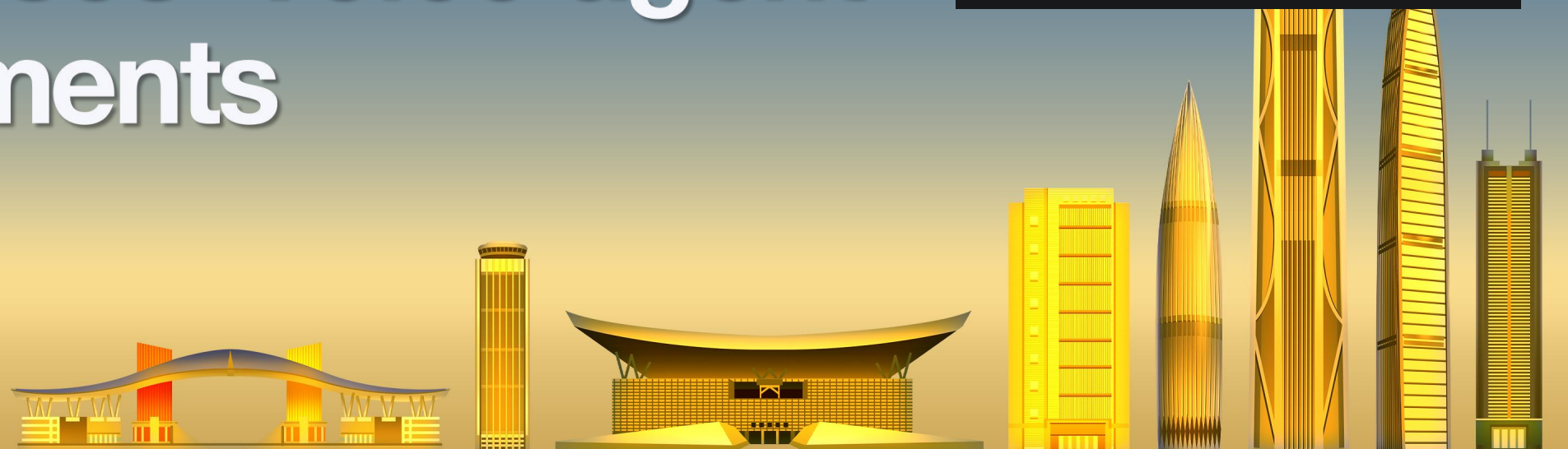
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02

Learnings from 100+ enterprises' voice agent deployments

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100+ enterprise voice agents observed

Where voice agents are creating real business value

Customer Service

Contact centers, telecom, customer care

Healthcare

Patient intake, appointments, follow-ups

Financial Services

Banking, insurance, collections

Retail & Hospitality

Orders, reservations, customer support

Sales & Recruiting

Lead qualification, outbound, candidate screening

Real Estate & Services

Property inquiries, automotive, field services

Education & Coaching

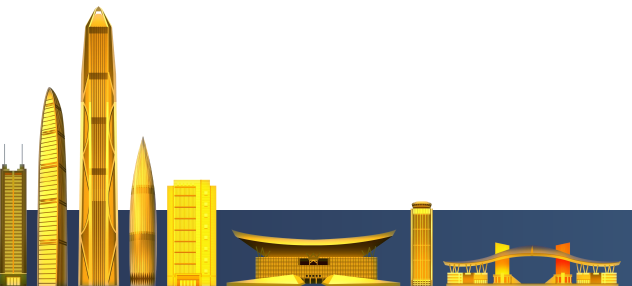
Language learning, role-play, training

Platforms & Infrastructure

Voice APIs, telephony, orchestration, evaluation

Our three product assumptions are heavily challenged

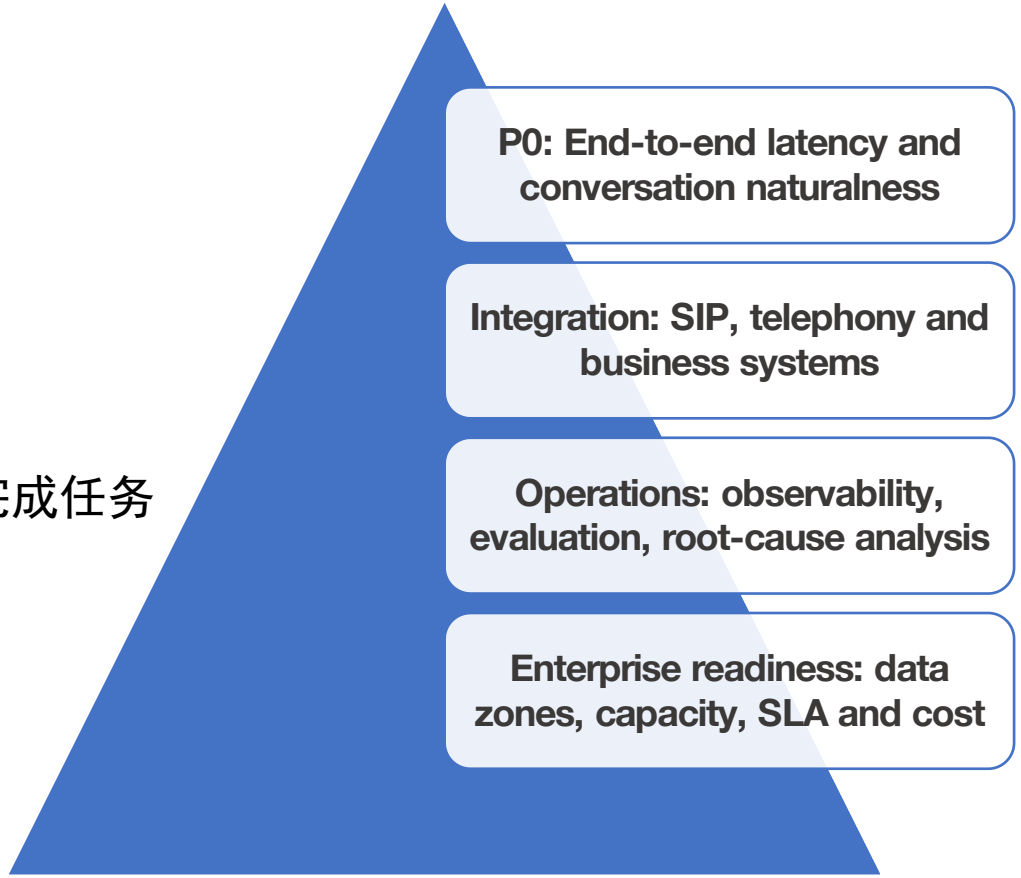
Better models would win	→	Customers demanded production-ready systems.
Speech quality was the hardest problem	→	Customers struggled with conversational timing and naturalness.
Unified APIs/services were enough	→	Customers needed complete platforms for the full voice agent life cycle.



What production customers actually need

The recurring blockers are system-level.

- 模型好 ≠ 对话好
- ASR 准 ≠ 用户理解正确
- TTS 像真人 ≠ 对话自然
- latency 低 ≠ 用户觉得快
- API 能跑 ≠ 产品能上线
- Agent 能回答 ≠ Agent 能完成任务



Latency is an end-to-end system property

The critical path includes capture, transport, models, tools, synthesis, playback, and recovery

CAPTURE	ASR	REASON + TOOLS	TTS	PLAYBACK
speech end	transcript	next action	first audio	useful response

USER-OBSERVED CLOCK: from speaking / interruption to a useful, context-correct continuation

TURN LATENCY

Report P50 / P95 end-to-end—not only average stage time.

INTERRUPTION STOP

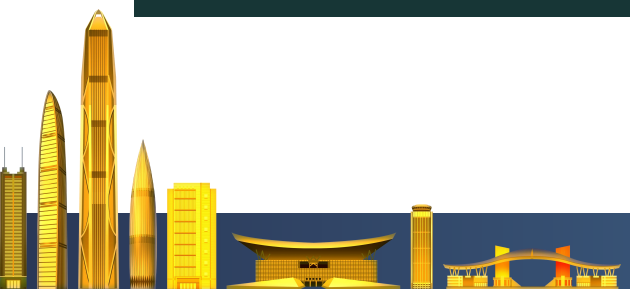
Measure how quickly output stops after the user barges in.

RECOVERY TIME

Measure time to restore state and continue usefully.

OPTIMIZE THE CRITICAL PATH

Use a latency waterfall to find the bottleneck by scenario—including network, SIP/telephony, queueing, orchestration, tools, model, and synthesis.

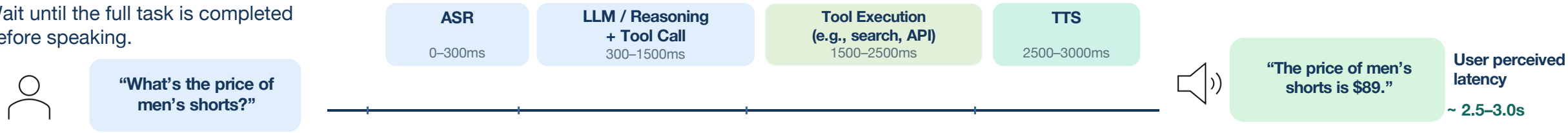


Use early speech and frontline chat to reduce user perceived latency

The agent can start speaking before the full task is completed, by using filler words or a front-end dialogue agent to provide immediate and progressive responses.

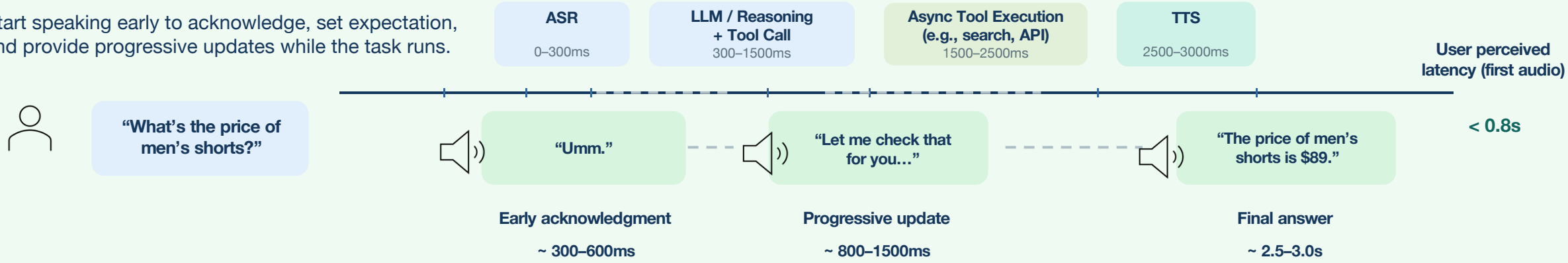
Standard response

Wait until the full task is completed before speaking.



With early speech

Start speaking early to acknowledge, set expectation, and provide progressive updates while the task runs.



REDUCE USER PERCEIVED LATENCY

A front-end dialogue flow or agent allow the system to speak early and progressively, reducing user perceived latency, even though the full task is still running in the background.

WHAT WE LEARNED

Strong WER/CER can still produce the wrong conversation

ASR accuracy is necessary — but a small recognition error can change the dialogue trajectory.

KEEP THE BASELINE

WER / CER

Still essential for:
- transcription accuracy
- model comparison
- regression diagnosis

$$\mathrm{WER} = \frac{S + D + I}{N}$$

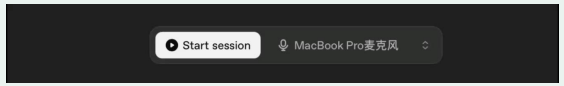
- **S**：替换错误数（Substitution）
- **D**：删除错误数（Deletion）
- **I**：插入错误数（Insertion）
- **N**：参考文本中的总词数（真值词数量）

REAL-WORLD FAILURE

LANGUAGE MISROUTING

“Hello” → German
Accent → Chinese

→ Agent switches language
→ Conversation breaks



WHAT WE LEARNED

CONTEXT MATTERS

Hard-code the language?
More predictable — less natural.

Use conversation context?
More adaptive — needs context-aware evaluation.

RECOMMENDED · WER/CER + language-routing errors + downstream conversation impact

WHAT WE LEARNED

Human-like TTS can still feel unnatural

Single-utterance realism does not guarantee conversational naturalness.

TRADITIONAL TTS EVALUATION

UTTERANCE-LEVEL MOS

- intelligibility
- pronunciation
- voice quality
- prosody
- MOS / human discrimination

REAL-WORLD OBSERVATION

THE VOICE SOUNDS HUMAN

Yet users still report:

- flat or misplaced emphasis
- wrong timing
- weak conversational flow
- responses that ignore context

THE MISSING LAYER

CONVERSATIONAL NATURALNESS

Does it:
- respond at the right moment?
- carry tone across turns?
- preserve context?
- adapt and recover naturally?

TTS naturalness is a property of an utterance. Conversational naturalness is a property of the interaction.

Conversation naturalness is about the full interactions

A NEW WAY TO EVALUATE

The user experiences the Agent as a conversation — not as a collection of speech components.

01 · SPEECH

Component Quality

WER / CER
MOS · voice quality

→

02 · CONVERSATION

Interaction Quality

turn-taking · timing
interruption · coherence · prosody

→

03 · OUTCOME

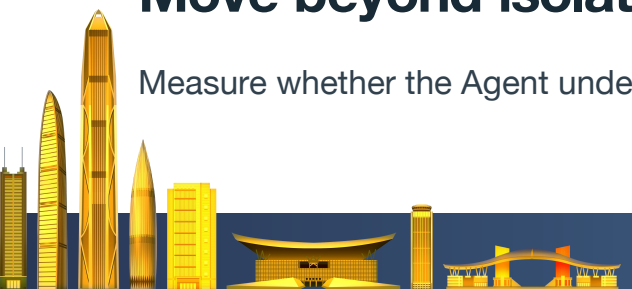
Agent Effectiveness

task success
recovery · user satisfaction

THE UNIT OF EVALUATION

Move beyond isolated component scores.

Measure whether the Agent understands, responds, adapts, and recovers across the full interaction.



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03

Moving towards Voice AI 2.0: The new voice-native agent paradigm

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From voice-enabled to voice-native

The difference is not whether the agent can speak — it is whether conversation shapes the system.

VOICE-ENABLED

Speech is an input/output layer around a primarily text-based agent.

TEXT AGENT + SPEECH

VOICE-NATIVE

Real-time speech conversation shapes the full lifecycle of the agent, from creation, deployment to operation.

Create — DEPLOY — OPERATE

Optimized for speech

KEY SHIFT

Speech is no longer just a channel; it becomes part of the agent's control loop.



CREATE — Design the conversation, not just the prompt

Build the interaction model before tuning the prompt

DEFINE STATES

Specify turn-taking, task boundaries, and when the agent should listen, speak, wait, or act.

DESIGN RECOVERY

Plan interruption, cancellation, clarification, and continuation paths.

USE CONTEXT

Disambiguate language and intent from conversation context without rigid constraints.

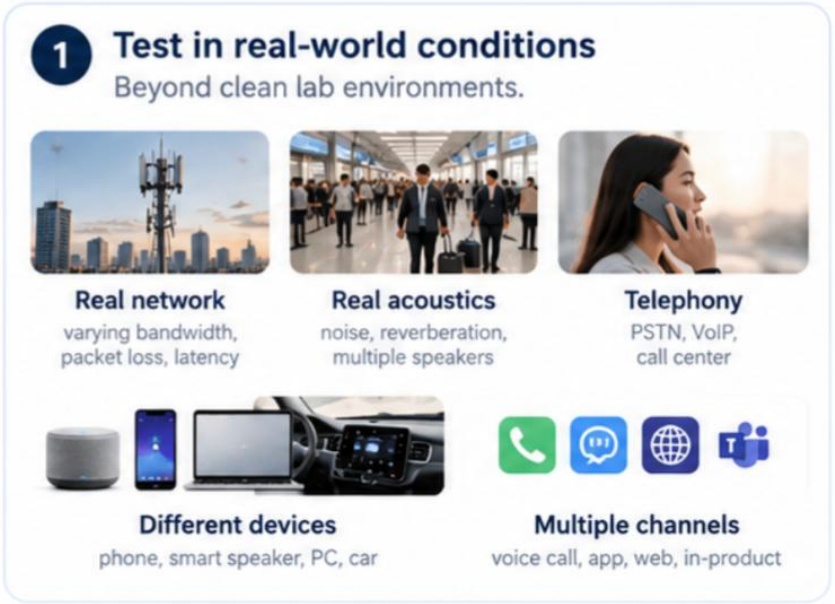
TEST EPISODES

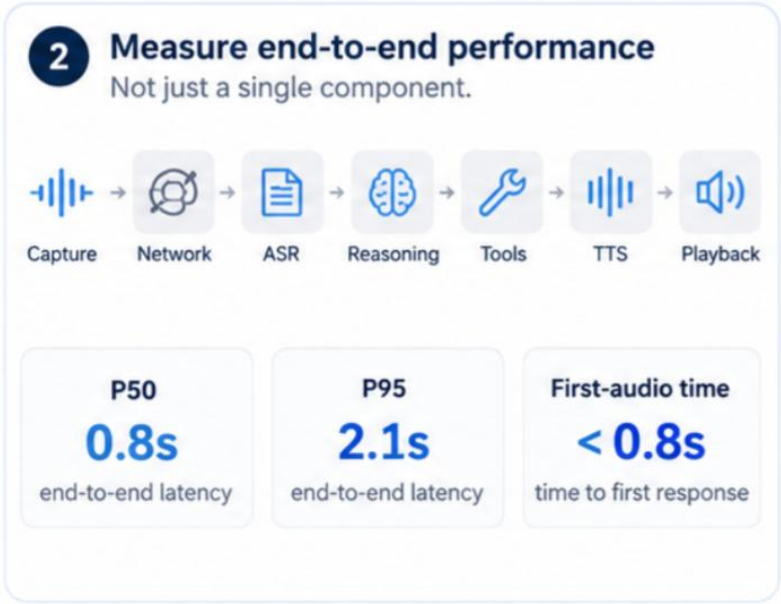
Evaluate multi-turn coherence, recovery, and task outcomes—not only isolated ASR/TTS clips.

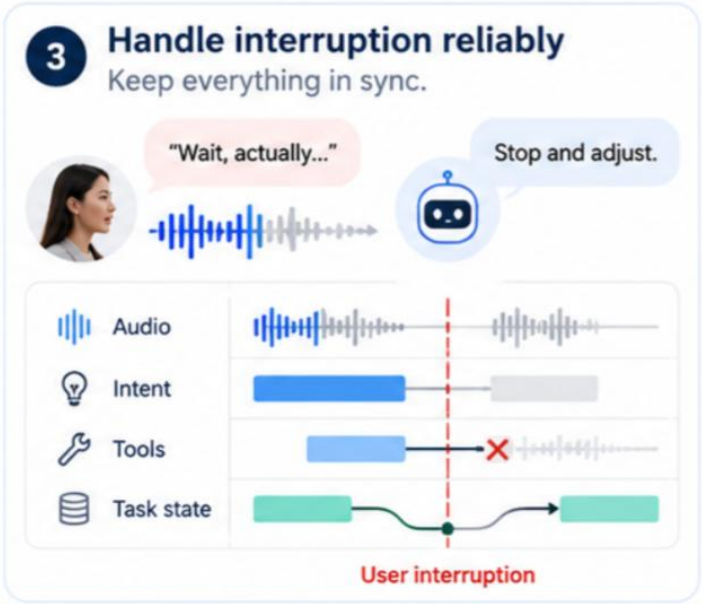


DEPLOY — Make the agent work in the real world

Validate the complete system under real operating conditions







TEST ENVIRONMENTS

Validate networks, acoustics, telephony, devices, and channel-specific behavior.

MEASURE LATENCY

Track end-to-end P50/P95 and first-audio time—not just individual component timing.

SYNC STATE

Keep audio, intent, tools, and task state aligned through interruption and recovery.

OPERATE — Observe, diagnose, and improve

Production conversations become evidence for the next iteration

RECONSTRUCT

Capture the conversation timeline, state transitions, tool events, and outcomes.

FIND PATTERNS

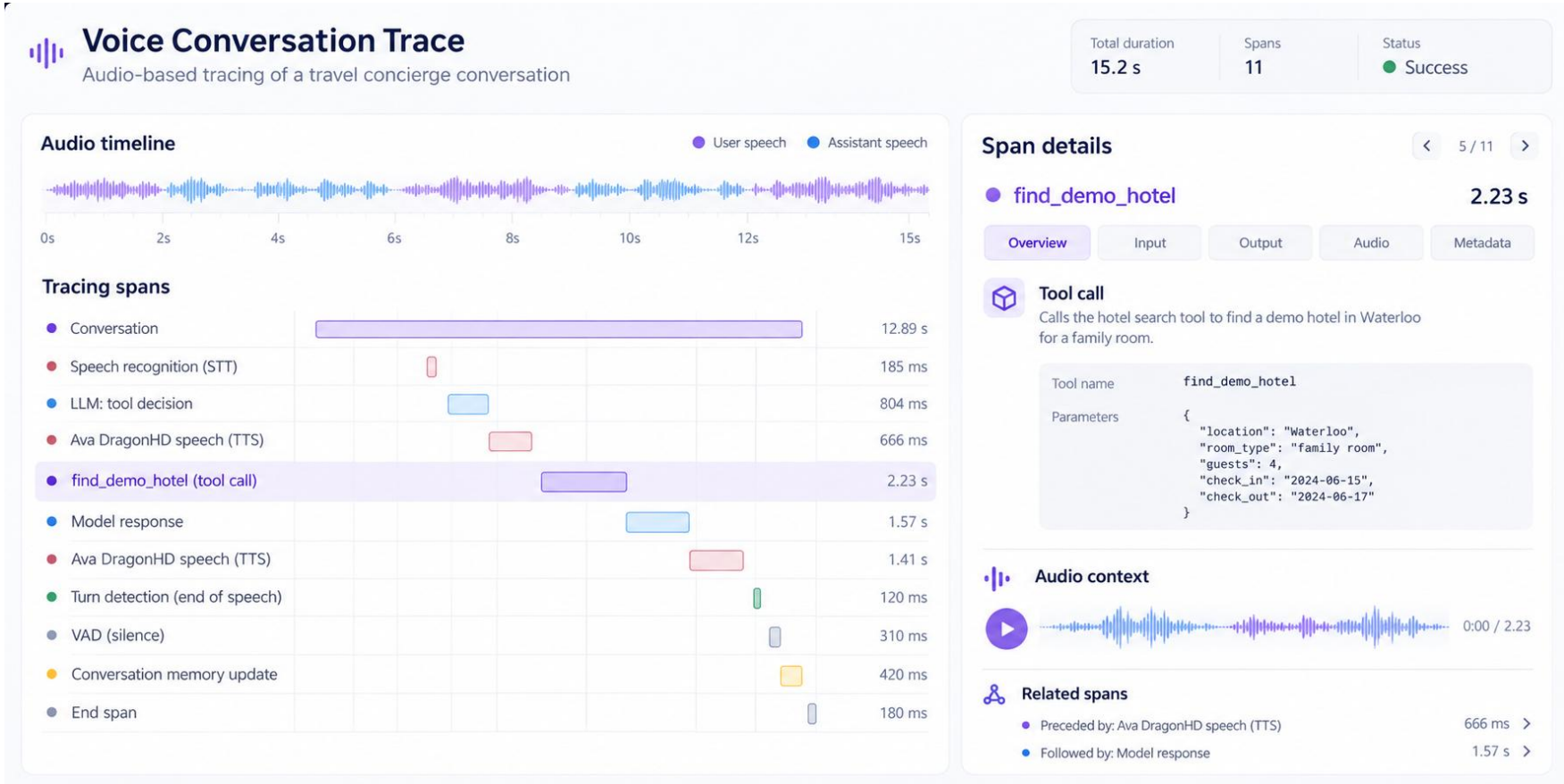
Segment failures by scenario, language, environment, and configuration.

DIAGNOSE

Connect user-visible failures to component and interaction-level root causes.

CLOSE THE LOOP

Convert incidents into regression tests and feed findings back into Create.



Building Voice-Native Agent at Ingora

A dual-model architecture that delivers real-time, natural conversation while getting complex tasks done.

Voice experience optimized for the full lifecycle

User Experience

Natural, engaging, and adaptive voice interaction for English learning.



Can you tell me a story about a brave little fox?

Sure! Let's start a story about a brave fox. Once upon a time...

Early acknowledgement
-0.6s

The fox faced a big challenge in the forest. What do you think it should do?

Progressive update
~ 0.8–1.5s

That's a great idea! You used "brave" and "clever" very well. Shall we continue the story?

Personalized response
~ 2.5–3.0s

Voice-Native Agent Core

Dual-model architecture for real-time conversation and complex task execution.

Orchestrator / Dialogue Manager

Turn-taking | Intent & task routing | Context management | Response planning

Audio Stream (from user)

Streaming Audio (to user)

Audio Interaction (Fast Response)

- Real-time conversational interaction
- Filler words and acknowledgement
- Short-form reasoning
- Natural and expressive speech output (TTS)

Task request (in parallel)

Partial / final result (streaming or final)

Task Completion (Long Task)

- Complex reasoning and planning
- Tool use (search, API, DB, code)
- Multi-step execution
- Longer running tasks
- Structured result output

Observability & Evaluation

Turn real conversations into insights for quality, safety and growth.

- **Latency & Turn Metrics** — Turn latency, interruption, recovery time
- **Conversation Quality** — Relevance, naturalness, user satisfaction
- **Task Success** — Completion rate, accuracy
- **Safety & Guardrails** — Content safety, privacy, alignment
- **Analytics & Learning** — Tracing, logs, evaluation, A/B testing, user insights

Ingora Intelligence

Four intelligences working together to make the agent understand, act and grow with each learner.

People Understanding Intelligence
Understands the learner, their context, goals and needs.
Learner Profile | Learning Goals | Memory

Capability Intelligence
Understands the task and how to use the right tools and skills to accomplish it.
Tool Use | Planning | Execution

Intervention Intelligence
Knows when and how to intervene for better learning outcomes.
Turn-taking | Scaffolding | Feedback

Growth Intelligence
Personalizes, adapts and grows with the learner over time.
Learning Path | Progress | Adaptation

Every conversation helps the learner learn better, build skills faster, and grow with confidence.

Empower everyone to grow with AI





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Thanks!

